

Mahesh Basavaraj

EDUCATOR · AI RESEARCHER

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TEACHING SNAPSHOT

8 years teaching CS

6+ courses taught

2 programmes designed

7 certifications & FDPs

APPROACH

First principles, case studies, hands-on — the why before the how, and a decision to make in every class.

COURSES TAUGHT

Operating Systems
UG · core

Data Structures & Algorithms
UG · core

Object-Oriented Programming (Java)
UG · core

Database Management Systems
UG · core

Deep Learning
PG · elective

Data Science & Analytics with Python
UG/PG · elective

Agentic AI (6-month immersion)
Industry-partnered

TEACHING TOOLS

Built in-house and used in class: CPU Rush (OS), Data Structure Design Studio, Game Forge MCP, ButterCloud Quiz, MLOps CI/CD lab. All public on GitHub.

Teaching Methodology & Philosophy

PREPARED FOR THE ASSISTANT PROFESSOR (COMPUTER SCIENCE) ROLE — UNSW BENGALURU CAMPUS

WHY I TEACH THE WAY I DO

Students rarely fail computer science because it is too hard. They fail because the first encounter is abstract: memory is a diagram, a scheduler is a paragraph, a model is a black box. My job is to make the invisible observable. When a student can predict what the machine will do, and then watch it do something else, the correction is theirs — and it stays. I design for that moment deliberately, and I build the software that makes it possible at class size.

The second half of the job is honesty about destination. Most of my students in Bengaluru will be employed as engineers within months of graduating. Theory has to be delivered in a form they can carry into a codebase — which is why my courses use real repositories, code review and delivery discipline, not only examinations.

My method is first-principles and case-study led. A topic opens with the basic necessity of the thing — why it exists, what breaks without it — and only then moves to how it works; the why earns attention, the how earns the skill. From there we work through case studies students can actually run, and wherever the subject allows, the learning is hands-on.

FIVE DESIGN PRINCIPLES

1. **Start from the misconception, not the syllabus.** Before each topic I write down the wrong mental model students usually hold, and design the first activity to expose it. Deadlock is taught by making them schedule two processes that lock each other out, not by listing Coffman conditions first.
2. **Make thinking visible, and therefore gradable.** Every class has decision points where students commit to an answer — predict, then observe. Commitments are captured and scored, so I can see whether the room actually understood before the exam tells me.
3. **Assess what matters, at the right stakes.** Low-stakes practice every week with instant feedback; high-stakes judgement reserved for a capstone that looks like real work. Rubrics are criterion-referenced and published in advance, so a grade means “can do X” rather than “was in the top 20%”.
4. **Treat generative AI as a professional instrument.** Banning it teaches nothing and is unenforceable. I permit it with mandatory disclosure, assess the reasoning through oral defence, and set tasks where an unverified generated answer fails on its own evidence — which is exactly the skill employers now need.
5. **Build for the whole room.** Every course ships with a progressive track: scaffolded entry points for students who need them, extension problems for those who do not, and no single bottleneck activity that only one learning style can pass.

PROFESSIONAL DEVELOPMENT

Deep Learning — Elite
NPTEL · Oct–Dec 2025

Blockchain Technology
NPTEL · Oct–Dec 2025

Data Analytics with Python
NPTEL · Jan–Apr 2024

Machine Learning & NLP
Infosys Springboard FDP · Jun 2025

Cyber Security
AICTE · VTU FDP · May 2023

Amazon Web Services
AICTE FDP · Aug 2022

The Science Behind Branding
IISc Bangalore · Dec 2025

TEACHING LEADERSHIP

- Designed and led a 6-month Agentic AI immersion programme with Capabl (industry partner).
- Organised and delivered institutional Faculty Development Programmes.
- Supervised final-year capstone teams to working artefacts and defence.

HOW I RUN A COURSE

Each week has the same rhythm, so students always know where the learning is being checked.

- **Before class —** a two-question diagnostic on the platform tells me the real starting point, not the assumed one.
- **In class —** live coding with deliberate errors to catch, a short prediction round, then peer instruction: commit, compare, defend.
- **Lab —** each lab has checkable gates. Attendance is not evidence; a passing gate is.
- **Weekly —** a short low-stakes quiz with immediate feedback, a worked best answer, and a cumulative score that rewards consistency.
- **End of course —** a capstone in a real repository with code review, a demo and an oral defence.
- **Every term —** anonymous mid-semester feedback, one visible change made in response, and one assessment redesigned.

EVIDENCE: TOOLS I HAVE BUILT AND USE

I do not only design on paper. These are working systems, public on GitHub, that carry the principles above into a live classroom.

CPU Rush Socratic OS game — students predict scheduling, paging, deadlock and file-system behaviour; a server grades every move.
Data Structure Design Studio Progressive scored decision track from arrays to hash maps, where constraints change and choices have consequences.
Game Forge (MCP) Generates single-file browser games, each teaching one concept through play.
ButterCloud Quiz Scenario platform: pick the best stack and architecture for realistic requirements.
MLOps CI/CD lab Production-style ML API where every push runs tests and model-quality gates.

ALIGNMENT WITH UNSW BENGALURU

My practice maps directly onto how UNSW teaches. I design at program level rather than course level, which is the logic of the Integrated Curriculum Framework: learning outcomes, activities and assessment aligned before any content is written. I am comfortable with standards-based, criterion-referenced assessment, with the myExperience approach to course design, and with work-integrated learning as an assessed component rather than an add-on — my industry years make the partner conversations realistic. I expect to learn a great deal from the Ready to Teach programme and from colleagues in Sydney and Canberra, and I am keen to contribute back: the teaching systems above are reusable, and I would happily adapt them for UNSW courses and share them across the campus network.

The new campus's AI Centre of Excellence with Deloitte India is a natural home for capstone work: students solving a partner's real problem, assessed on both engineering quality and research rigour. That is the kind of assessment I want to build here.

FIRST TWELVE MONTHS

- **Complete Ready to Teach —** and adopt myExperience and the Integrated Curriculum Framework in every course I take on.
- **Publish aligned course outlines —** with constructive alignment from program learning outcomes down to each assessment task.
- **Introduce one industry-linked assessment —** per course, co-designed with a partner and marked against a published rubric.
- **Contribute a teaching case study —** on AI-era assessment integrity to the Bengaluru and Sydney teaching communities.
- **Support the AI Centre of Excellence —** by connecting capstone projects to its research themes and supervising them to publication quality.